OCP SPECIAL FOCUS:
PHOTONICS



µLED Interconnects:
Addressing the Future of Scale-In and Scale-Up Connectivity

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AI scales in four directions

Scale-across

100s to 1000s of kilometers

Data center to data center
Across multiple data centers

Scale-out

Meters to kilometers

Server to server
Within data center

Scale-up

Up to 10 meters for inference
Up to 50 meters for training

XPU to XPU
Within rack or row

Scale-in

Millimeters to Meter

Chip to chip, chip to memory
Within package or PCB

Optical today

Copper today

Why copper wins today for scale-up and scale-in

Reliability



Power



Latency



Heat tolerance



Simplicity



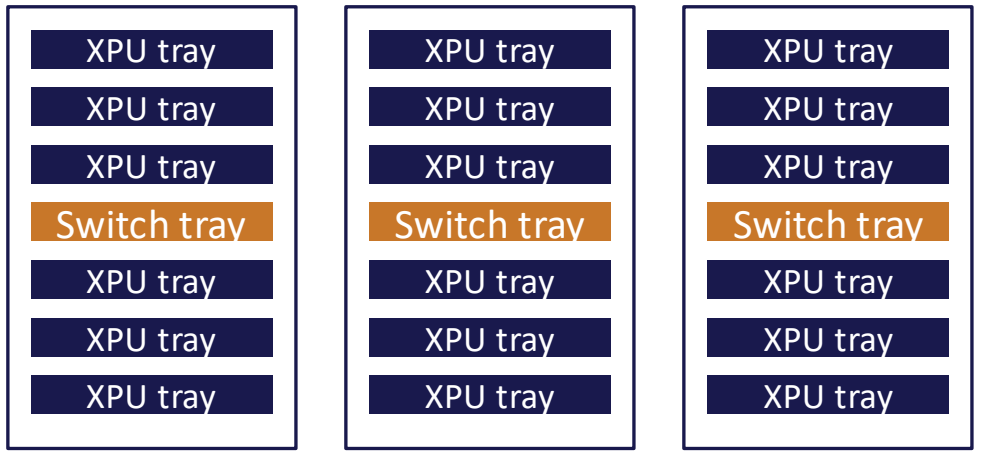
Cost



Copper when you can, optics when you must

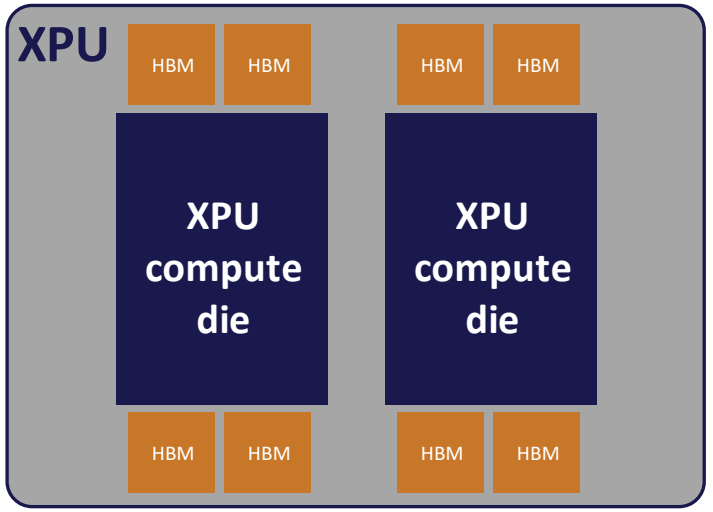
Copper hitting walls to enable required bandwidth scaling

Scale-up



Reach, power, latency, weight and size

Scale-in



Reach, routing density, power and latency

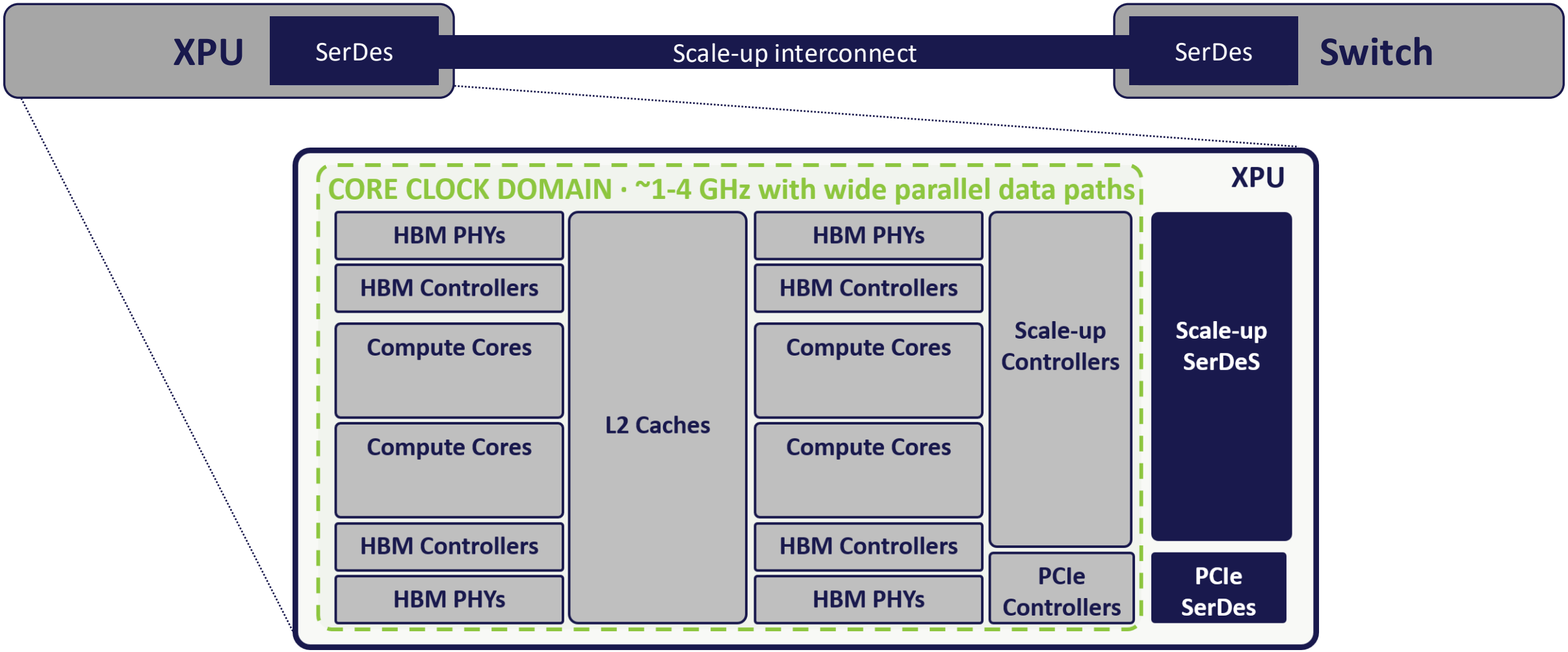
Significant opportunity for optical connectivity but….

Current optics **do not meet** the non-reach requirements

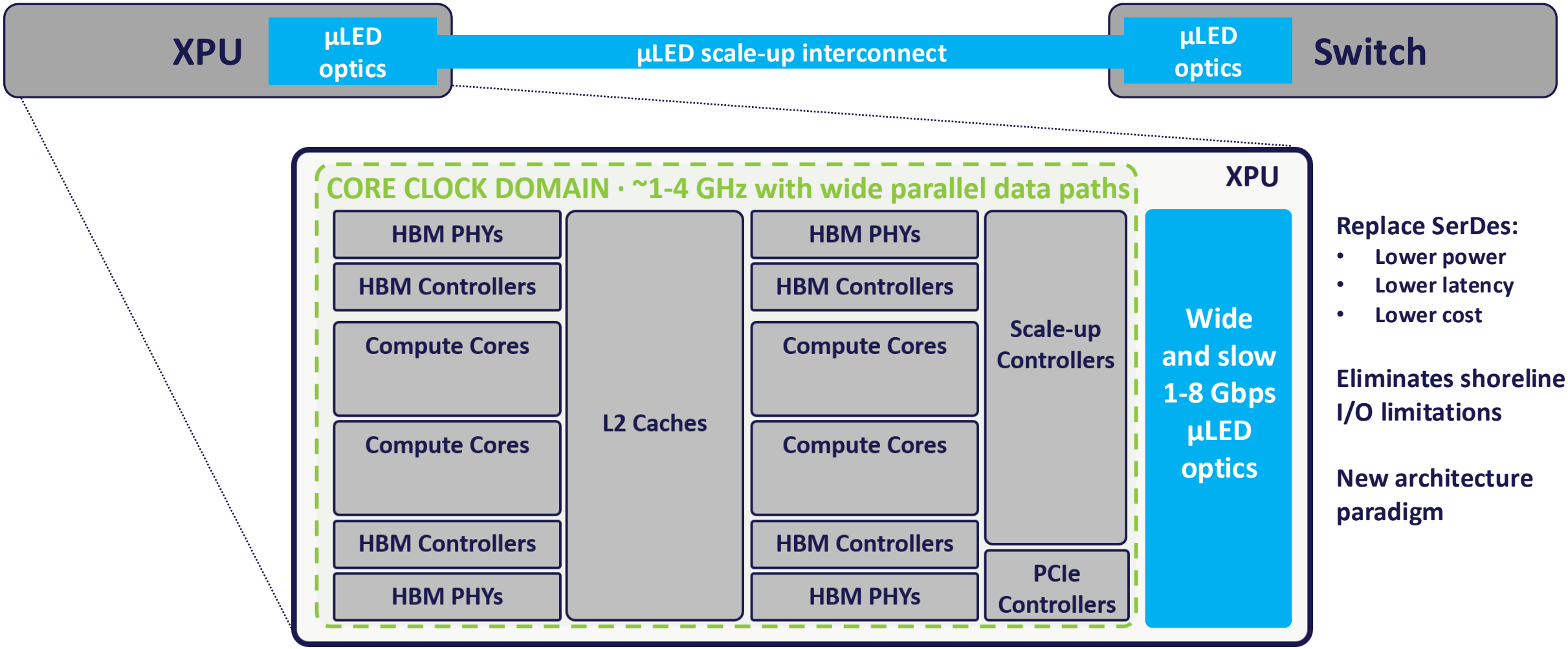
	Current optical solutions
Scalable high bandwidth density	Shoreline-limited
High reliability	Laser-limited – complex, costly subsystem needed
Low power (high energy efficiency)	>10pJ/bit (includes >5pJ bit host SerDes)
Near-zero latency	DSP retiming and higher-order modulation
High temperature tolerance	Laser-limited – complex, costly subsystem needed
Simple silicon integration	Multiple non-silicon components needed
Low cost	>>>$0.10 per Gbps

Need optical system solution that approaches or exceeds copper parity

Current scale-up end-to-end interconnect architecture



µLEDs enable native end to end wide and slow architecture



Replace SerDes:

- **Lower power**
- **Lower latency**
- **Lower cost**

Eliminates shoreline I/O limitations

New architecture paradigm

µLED interconnects tackle scale-up and scale-in requirements

	µLED interconnects
Scalable high bandwidth density	>10Tbps/mm shoreline and >10Tbps/mm^2 area
High reliability	No lasers and array spare lanes redundancy
Low power (high energy efficiency)	<1pJ/bit with no threshold current
Near-zero latency	NRZ native: no SerDes, DSPs and PAM4/PAM6/etc
High temperature tolerance	>200°C
Simple silicon integration	Wafer-level processing with simple transfer processes
Low cost	<$0.10 per Gbps (no lasers, DSPs, modulators/etc)

Exceeds copper parity at end-to-end system level

Single interconnect architecture for scale-up and scale-in

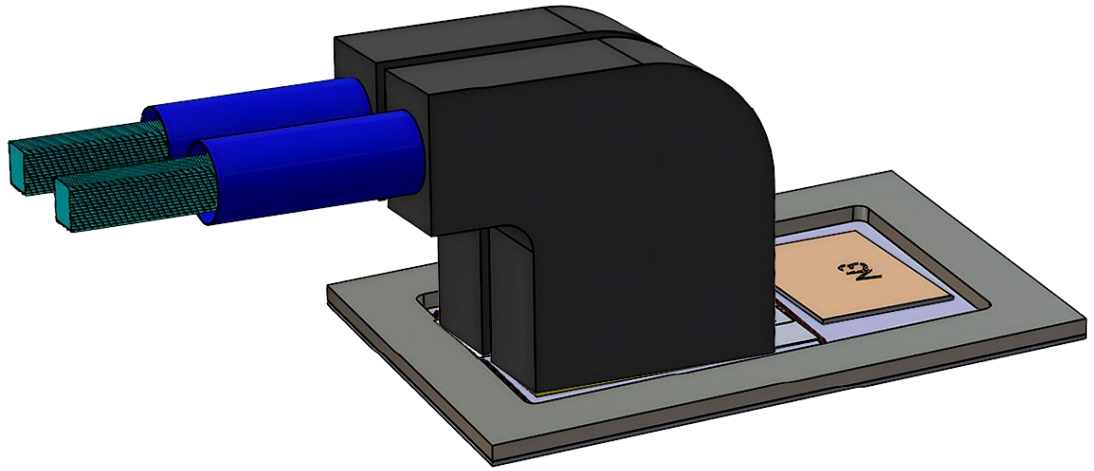
Scale-up (copper replacement up to 10m)



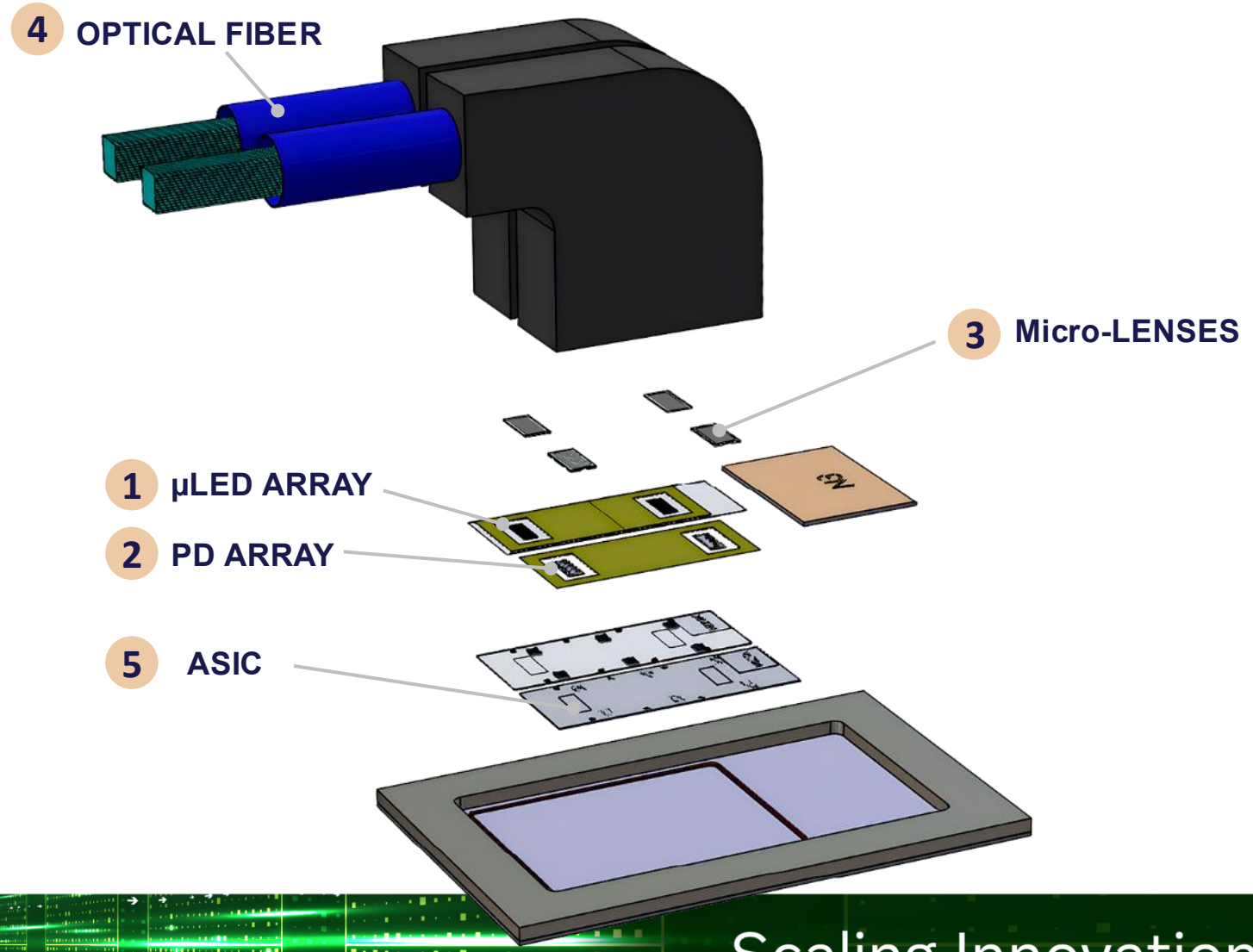
Scale-in



Avicena pioneering µLED interconnects for AI infrastructure

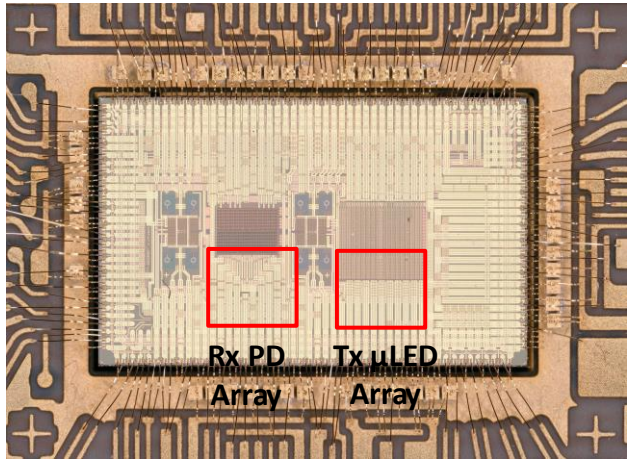


Avicena pioneering µLED interconnects for AI infrastructure



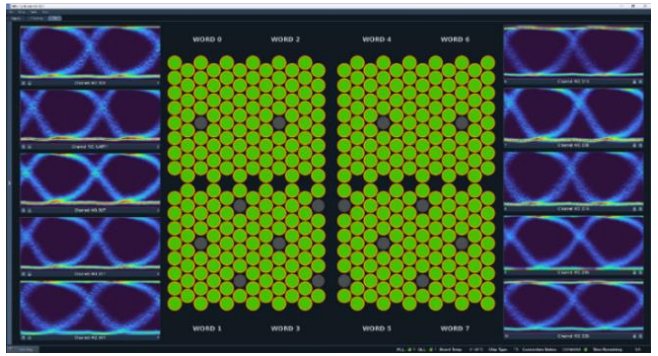
Shipping world's 1st µLED interconnects: 1Tbps ASIC-powered

LightBundle™ 1Tbps ASIC
Integrates 300+ drivers and TiAs with µLED, PD and microlens arrays



~1mm² of µLED array for >1Tbps of bandwidth

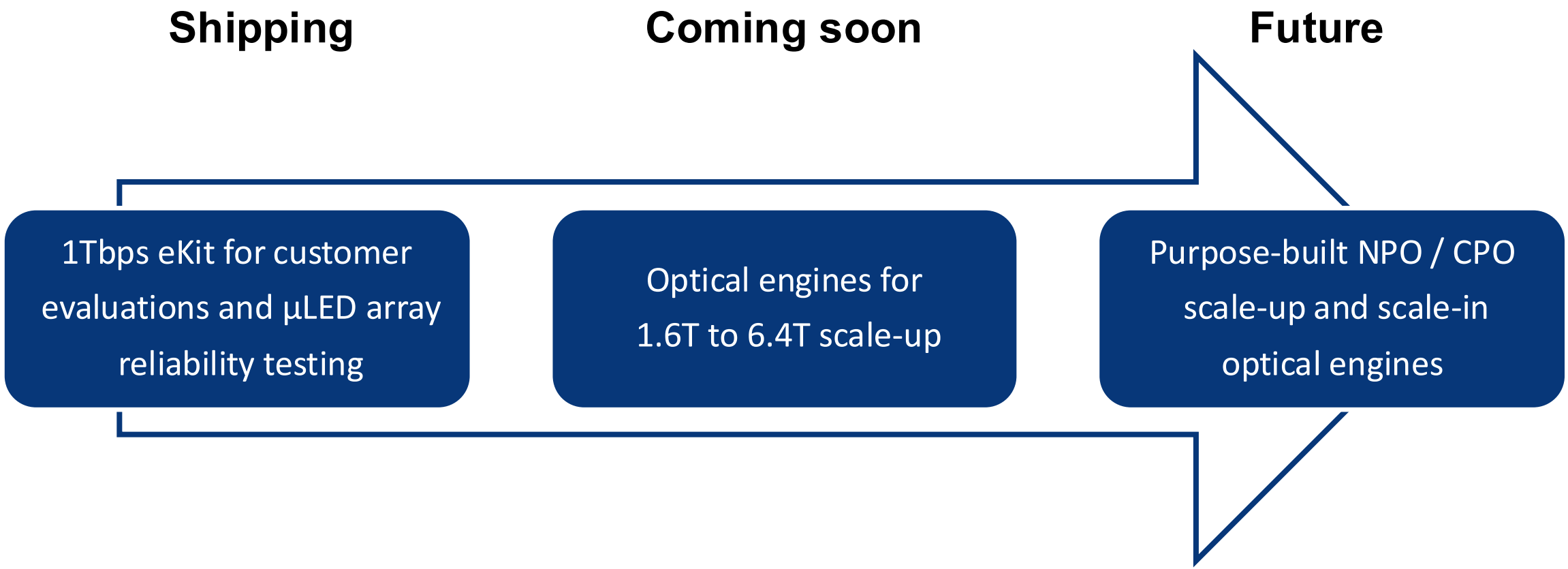
LightBundle™ 1Tbps eKit
Array of 335 µLED channels each operating at 3Gbps





Customers and partners testing in their labs

Avicena µLED interconnect product roadmap plan



Key takeaways and call to action

1. Copper hitting multiple walls for next generation scale-up and scale-in connectivity

2. Current optics **do not meet** scale-up and scale-in requirements

3. µLEDs enable disruptive slow and wide interconnect architecture for scale-up and scale-in

Call to action:

Collaborate at OCP short-reach workstreams to define requirements and standards

Thank You!

